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ANOMALY DETECTION WITH ANIMAL SENSITIVITY

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I. ABSTRACT

Animals have been observed for centuries to exhibit unusual behavioural and physiological responses prior to certain environmental disturbances such as earthquakes, extreme weather events, and ecological stress conditions. These instinctive reactions are believed to arise from animals' heightened sensitivity to subtle environmental changes that may not be immediately detectable by conventional monitoring systems. However, translating such observations into a structured, scalable, and automated analytical framework remains a significant challenge.

This research presents an AI-based anomaly detection framework for analysing animal behavioural data to identify abnormal patterns that may act as early indicators of environmental disturbances. The study utilises multi-species datasets, including elephant movement and temperature data as well as dog activity and physiological summaries. Key behavioural features such as movement speed, acceleration, directional changes, activity intensity, and temperature variations are extracted and analysed.

Rather than directly predicting natural disasters, the proposed approach focuses on unsupervised learning to model normal animal behaviour and detect deviations from learned behavioural baselines. Generative AI techniques are optionally incorporated to address the scarcity of rare behavioural events by synthesising representative anomaly-like data. Experimental analysis demonstrates that meaningful behavioural anomalies can be detected without the need for labelled disaster data. This research contributes towards the development of biologically informed, data- driven early warning support systems that complement existing environmental monitoring technologies.

Keywords: *Animal Behaviour Analysis, Anomaly Detection, Unsupervised Learning, Environmental Monitoring, Generative AI, Early Warning Systems, LSTM.*

II. INTRODUCTION

Natural disasters such as earthquakes, floods, and extreme climatic events continue to pose significant risks to human populations, infrastructure, and ecosystems. Early detection and preparedness play a crucial role in minimizing damage and loss of life. Traditional disaster prediction and monitoring systems rely heavily on physical sensors, including seismographs, meteorological stations, satellites, and ocean buoys. While these systems provide valuable data, they are often expensive, infrastructure-intensive, and sometimes limited in their early warning capabilities.

Animals possess highly sensitive biological and neurological mechanisms that allow them to perceive subtle changes in their environment. Numerous anecdotal reports and scientific studies suggest that animals may respond to environmental disturbances earlier than humans. These responses may include unusual movement, restlessness, migration changes, or physiological stress reactions. Wikelski et al. demonstrated that farm animals exhibit measurable behavioural changes prior to earthquakes, indicating the potential of animal behaviour as an early warning signal [1].

With recent advancements in artificial intelligence and the availability of large-scale animal behavioural datasets, it is now possible to systematically analyse such behaviour patterns. This research aims to explore the use of AI-driven anomaly detection techniques to identify abnormal animal behaviour that may support environmental disturbance awareness.

III. PROBLEM DEFINITION

Direct prediction of natural disasters using machine learning techniques is inherently challenging. Disaster events are rare, unpredictable, and difficult to label accurately in historical datasets. Furthermore, animal behavioural data is continuous, noisy, and influenced by multiple biological, ecological, and environmental factors. These challenges make supervised learning approaches impractical for this domain.

The core problem addressed in this research is the lack of an automated, scalable framework capable of detecting abnormal animal behaviour without relying on explicit disaster labels. There is a need for an unsupervised system that can learn normal behavioural patterns and identify deviations that may indicate environmental instability. Such a system should be adaptable to multiple species, robust to noise, and suitable for real-world deployment.

IV. SIGNIFICANCE OF ANIMAL BEHAVIOURAL ANALYSIS

Animal behaviour offers a biologically grounded source of environmental information. Elephants, for example, have been observed to alter their movement patterns, migration routes, and physiological responses under environmental stress conditions [1]. Dogs, which are commonly monitored using wearable sensors, exhibit changes in activity intensity, heart rate, and behaviour during abnormal conditions, as shown in recent datasets [2].

Analysing behavioural anomalies provides an additional layer of information that can complement traditional environmental monitoring systems. Rather than replacing physical sensors, animal behaviour-based analysis can serve as an early supporting signal, particularly in remote or ecologically sensitive regions where conventional infrastructure may be limited. This biologically informed approach enhances situational awareness and contributes to a more holistic understanding of environmental dynamics.

V. ROLE OF ARTIFICIAL INTELLIGENCE IN BEHAVIOURAL ANALYSIS

Artificial intelligence enables the automated analysis of high-dimensional, time-series behavioural data. Unsupervised anomaly detection techniques are particularly well-suited for this research, as they do not require labelled outcomes. According to Chandola et al., anomaly detection methods are effective in identifying rare and unusual patterns in complex datasets [4].

By learning normal behavioural baselines, anomaly detection models can flag deviations that may correspond to abnormal environmental conditions. Generative AI models further enhance this process by addressing data imbalance. Xu and Deng demonstrated that generative models can synthesise realistic time-series data, improving robustness and generalisation in rare-event detection scenarios [5]. This research leverages these techniques to strengthen behavioural anomaly detection performance.

VI. LITERATURE REVIEW

Several studies have investigated the relationship between animal behaviour and environmental disturbances. Wikelski et al. [1] provided empirical evidence that farm animals exhibit measurable behavioural changes prior to earthquakes, supporting the hypothesis that animals respond to environmental cues before physical events are detected by instruments.

Lee et al. [2] introduced a public dataset of dog vital signs collected using ultra-wideband radar technology, enabling detailed behavioural and physiological analysis. Chandola et al. [4] conducted a comprehensive survey of anomaly detection techniques, highlighting the suitability of unsupervised approaches for scenarios with limited labelled data. Xu and Deng [5] explored generative models for synthetic time-series data augmentation, addressing challenges related to data scarcity.

Additionally, findings presented in the Symposium on Ethology and Environmental Forecasting

[3] emphasize the importance of integrating animal behaviour into environmental monitoring frameworks. Despite these contributions, most existing research focuses on single species or observational analysis. This research addresses the gap by proposing a unified, AI-driven, multi- species anomaly detection framework.

Recent research has also highlighted the growing role of computational intelligence in transforming traditional ethological studies into scalable analytical systems. The findings presented in the *Symposium on Ethology and Environmental Forecasting* emphasize that continuous animal monitoring, when combined with data-driven models, enables objective detection of behavioural deviations that may be overlooked through manual observation alone [3]. Furthermore, anomaly detection techniques surveyed by Chandola et al. demonstrate strong suitability for such applications, particularly in scenarios where labelled event data is unavailable or incomplete [4].

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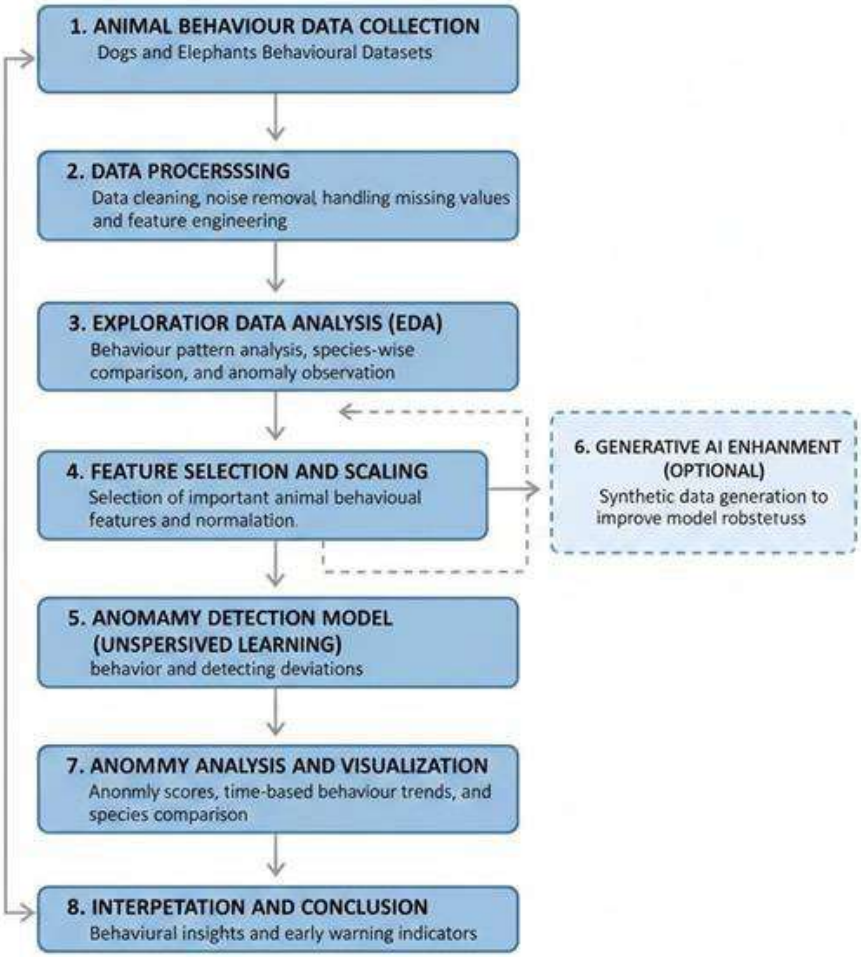
VII. METHODOLOGY

Workflow :



The proposed methodology consists of the following stages:

1. **Data Collection:** Behavioural and physiological datasets are collected from multiple animal species, including dogs and elephants.
2. **Data Preprocessing:** Raw data is cleaned, noise is removed, missing values are handled, and relevant behavioural features are extracted.
3. **Exploratory Data Analysis:** Behavioural patterns and variations are analysed to establish normal behaviour baselines.
4. **Feature Selection and Scaling:** Important behavioural indicators are selected and normalised to ensure balanced learning.
5. **Anomaly Detection Model:** An unsupervised learning model is trained to learn normal behaviour and detect deviations.
6. **Generative AI Enhancement:** Synthetic behavioural data is generated to improve model robustness.
7. **Anomaly Analysis:** Detected anomalies are visualised and interpreted over time.

VIII. PROPOSED SYSTEM

The proposed system functions as an automated behavioural anomaly detection pipeline. Animal behavioural data is processed through multiple analytical layers to generate anomaly scores representing deviations from normal behaviour. These scores act as early behavioural indicators of potential environmental disturbances.

The system is designed to support, rather than replace, existing disaster monitoring technologies. Its modular architecture allows future integration with additional data sources such as meteorological or seismic information.

IX. CONCLUSION

This research demonstrates that AI-based anomaly detection techniques can effectively identify abnormal animal behaviour without requiring labelled disaster data. By analysing multi-species behavioural patterns, the proposed framework provides a biologically informed approach to environmental disturbance awareness.

The findings support the hypothesis that animals exhibit detectable behavioural deviations prior to environmental changes. The proposed system offers a scalable, ethical, and scientifically responsible method for enhancing early warning support systems. Future work may focus on real- time deployment, integration with physical sensor data, and validation across additional animal species.

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